

DAEN 427/ISEN 427 Decision and Risk Analysis

Fall 2025

Instructor: Alexander Abuabara
Email: abuabara@tamu.edu

Time: T-R 9:35–10:50
Place: ETB 1013

Instructor Office Hours: Wed 1-2pm at <https://tamu.zoom.us/j/9792099780> or by appointment.

Prerequisites: Grade of C or better in ISEN 310, DAEN 321, or STAT 212.

Cross Listing: ISEN 427/DAEN 427.

Description: Overview of the state of the art in descriptive and prescriptive theories of decision making under uncertainty with emphasis on the ways in which human decisions depart from normative models of rationality; analytical foundations stemming from several disciplines, economics, psychology, management science; application in engineering systems will be considered.

Objectives: After taking this course, students will have the ability to:

1. Understand Bayesian approach to perception and decisions under uncertainty.
2. Implement computational models for Bayesian inference.

Textbooks: (recommended)

- *Bayesian Models of Perception and Action: An Introduction.* W. J. Ma, K. Kording, D. Goldreich (2022)
- *An Introduction to Bayesian Inference, Methods and Computation.* N. Heard (2021)
- *A Course in Behavioral Economics.* E. Angner (2021)
- *Bayesian Modeling and Computation in Python.* O. A. Martin, R. Kumar, J. Lao (2021)

Assignments & Grading Policy:

Homework/Exercises 20%; 2x Mid-term Exams 25% each; Final Project 30%

All assignments are expected to be completed individually unless otherwise specified. Any disagreements regarding a grade received on any graded material must be discussed within one week of the return of the graded material. No grade will be changed beyond the one week limit. If you request a regrade, the entire assignment will be regarded (i.e. if we made a mistake in your favor, this will also be corrected). Grades will be calculated on the basis of total points earned. If deemed necessary, the grade thresholds may be curved at the instructor's discretion. The thresholds for letter grades are:

$A \geq 90$, $B \geq 80$, $C \geq 70$, $D \geq 60$, $F < 60$.

Attendance Policy: You are expected to attend all class lectures except for university excused absences. The university rule regarding excused absences can be found at <http://student-rules.tamu.edu/rule07>.

Late Work: Late work will not be accepted except for approved university excused absences.

Office Hours: We are usually in our offices during office hours and are glad to discuss anything with you except material covered during an unexcused absence. If other students are already in our offices, please join them. If you would like to discuss a private matter with us please schedule an appointment.

Academic integrity: follows the Texas A&M Student Honor Code; all code notebooks and documents scanned with plagiarism detector. Don't cheat! If your temptation is to not turn in work that isn't yours, don't do it. Come see me instead and ask questions, I'm here to help. Content is cumulative, so the earlier, the better. You are going to have to write codes in your future job interviews, so learn it now!

Academic Honesty: The Industrial & Systems Engineering Faculty uphold the Aggie Honor Code as an axiom of academic excellence. We consider its sincere observance to be essential for membership in our department and Texas A&M. We extend you the trust conferred to those who faithfully adhere to our honor code. Abuse of this trust is intolerable, thus we will report and assign an extreme penalty to those who do not stand with us in preserving the integrity symbolized by the Aggie Honor Code: "An Aggie does not lie, cheat, or steal or tolerate those who do."

In this course, we will request the Aggie Honor Office to assign a grade of F for any violation of the Aggie Honor Code, no matter how minimal the violation may be. The Honor Council Rules and Procedures can be found at <http://aggiehonor.tamu.edu>

It is acceptable for you to discuss any assignments except exams with other students prior to the due date. After discussing an assignment with someone else, do not write anything that you do not understand and always use your own language on the assignments. Copying another student's writing directly, even if it is a portion of the assignment, is considered cheating and an Aggie Honor Code violation. No leniency is given for academic dishonesty.

The Americans with Disabilities Act (ADA): A federal anti-discrimination statute that provides comprehensive civil rights protection for persons with disabilities. Among other things, this legislation requires that all students with disabilities be guaranteed a learning environment that provides for reasonable accommodation of their disabilities. If you believe you have a disability requiring an accommodation, please contact Disability Services, currently located in the Disability Services building at the Student Services at White Creek complex on west campus or call 979-845-1637. For additional information visit <http://disability.tamu.edu>

Statement on Diversity and Inclusion: It is important to us that all students feel accepted and comfortable in this class. If for any reason you feel disrespected, uncomfortable, or unsafe in our classroom please discuss your concerns with us via the medium you are most comfortable with. If you do not feel comfortable communicating directly to us, please reach out to Dr. Ntamo (department head).

Other Pertinent Course Information: All information related to the course, including handouts, will be made available on <https://canvas.tamu.edu>. All communication related to the course will be sent electronically via Canvas as announcements or delivered to your @tamu.edu mailbox. You are expected to check your email regularly.

Course Outline: (tentative)

Lecture	Topic
1	Perception as inference. Bayesian inference
2	Bayesian inference under measurement noise
3	Uncertainty and confidence
4	Predicting behaviors, estimation (MLE, MAP)
5	Cue combination, learning as inference 1st midterm exam (in class)
6	Computational aspects of Bayes inference (MCMC)
7	Discrimination and detection
8	Binary classification
9	Bayesian linear regression I
10	Bayesian linear regression II 2nd midterm (in class)
11	Combining inference with utility
12	Rational choice under risk
13	Some paradoxes, prospect theory
14	Expectation maximization algorithm
15	Variational inference
16	Bayesian inference with neural networks Projects (in class)